ALGORITHM COMPLEXITY EXERCISE
DISCRETE MATHEMATICS

Lecturer :

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1. How many addition operations are performed in the following algorithm?

```
for i ← 1 to n do
    for j ← 1 to n do
        for k ← 1 to j do
            x ← x + 1
        endfor
    endfor
endfor
```

The answer :

$x \leftarrow x + 1$

The innermost loop runs = j times

The second loop runs = from 1 to n

The outermost loop runs = n times

Therefore

$$T(n) = \sum_{i=1}^{n} \sum_{j=1}^{n} \sum_{k=1}^{j} 1$$

Since the innermost loop executes j times :

$$T(n) = \sum_{i=1}^{n} \sum_{j=1}^{n} j$$

we know $\sum_{j=1}^{n} j = \frac{n(n+1)}{2}$ substitute $T(n) = n \cdot \frac{n(n+1)}{2}$

Then final answer $T(n) = \frac{n^2(n+1)}{2}$

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2. Count the number of multiplication and addition operations performed by the algorithm polynomial $P(x) = a0 + a1x + a2x^2 + ... + anx^n$

```
Function p(input x: real) → real
{ Returns the value of p(x) }

Dictionary
  j,k : integer
  jumlah, suku : real

Algorithm
  jumlah ← a0

  for j ← 1 to n do
    suku ← aj
    for k ← 1 to j do
      suku ← suku * x
    endfor

    jumlah ← jumlah + suku
  endfor

  return jumlah
```

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A. Multiplication Operations

The multiplication operation is

sum ← sum * x

For every j, the inner loop runs j times. Therefore $T(n) = \sum_{j=1}^{n} j$

Using the arithmetic series formula $T(n) = \frac{n(n+1)}{2}$

Total multiplication operations $\frac{n(n+1)}{2}$

B. Addition Operations

The additions operation is:

jumlah ← jumlah + sum

This statement executes once for each value of j.

Since j runs from 1 to n:

$T(n) = n$

Total Addition Operations

n

3. Determine the equation T(n)

```
Function looploop(n) {
    for (let i = 1; i <= n; i *= 2){
        For (let r = 1; r <= i; r++){
            for (let k = 1; k <= n; k++){
                console.log ("Matdis Seru!");
            }
        }
    }
}
```

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The statement console.log ("mardis sen!"); is the main operation counted. Outer Loop for (let i = 1; i <= n; i *= 2)
values of i: 1, 2, 4, 8, ...2^m where $2^m \leq n$.
Middle Loop For (let r = 1; r <= i; r++) runs i times.
Inner Loop for (let k = 1; k <= n; k++) runs n times.
Therefore $T(n) = n(1 + 2 + 4 + 8 + \ldots + 2^m)$
Using the geometric Series Formula:

$$1, 2, 4 + \ldots + 2^m = 2^{m+1} - 1$$

Since $2^m \approx n$ then $T(n) = n(2n - 1)$

$$T(n) = 2n^2 - n$$

The final Answer time complexity $O(n^2)$

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